

Predicting Song Release Decades Using Spotify Audio Features and Lyric Complexity

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Abstract—This interim report investigates the *SpotGenTrack* dataset from both the machine learning and hypothesis-testing perspectives. The first research question seeks answers regards to whether it is possible to predict the decade of a song that was released by using the Spotify audio features and the complexity of lyrics under the scope of a multi-class classification. The second research question investigates whether the old (songs that were released in or before 1980) and modern songs (released in or after 2010) are differentiating from each other based on their mean “acousticness” and “energy” values. In order to prepare for these analyses, the *SpotGenTrack* dataset was prepared through the cleaned and merged tables that include information regarding the tracks, albums, artists and lyrics. On the chosen variables, data profiling, missing value analysis, descriptive statistics and exploratory visualizations were conducted. In addition, an initial model and appropriate evaluation metrics for the machine learning iteration were defined. The findings of this interim report create a reproducible basis for the analyses to be conducted on the final part of this project.

Keywords—music information retrieval, multi-class classification, machine learning, data profiling, hypothesis testing, exploratory data analysis

I. INTRODUCTION

The rapid growth of music streaming platforms, such as Spotify, Apple Music, and SoundCloud etc. and digitalization have made the analysis of large datasets in the field of Music Information Retrieval (MIR) increasingly important. This project aims to examine the release decades of songs through both auditory and textual characteristics, using the *SpotGenTrack* dataset that contains 101,939 rows of audio features, lyrics, and metadata that were collected from Spotify and Genius, as the name of the dataset speaks for itself [1].

The literature contains many important studies demonstrating the success of combining audio and text features in machine learning and classification problems. For instance, Oramas et al. showed that multimodal deep learning approaches combining audio, text aka song analyses, and image representations significantly improved classification accuracy in music genre classification [2]. Similarly, Mayer et al. performed genre classification in digital music collections by combining text statistics and word features derived from lyrics with acoustic sound characteristics and proved that combining different data models increased success [3]. The power of this multimodal approach has also been proven in “mood classification” in music; Hu and Downie achieved a significant performance increase compared to systems using only sound by combining lyric features (linguistic and formal) with musical sound features and found that this combination reduced the amount of training data the model needed [4]. In

another very recent study, Nofer et. al used their “DeepHits” model, also developed on the *SpotGenTrack* dataset, to perform multi-class classification by combining Spotify audio features, lyric (BERT) embeddings, and metadata such as release year [5]. This study clearly demonstrates that the use of metadata and text features in conjunction with acoustic features plays a critical and transformative role in classification success [5].

Finally, Martín-Gutiérrez et. al, who introduced the *SpotGenTrack* dataset used in this project to the literature, proposed a multimodal end-to-end deep learning architecture that predicts song popularity using audio features, metadata, and lyric features obtained from Spotify and Genius platforms [1]. Considering these studies, the following research questions have been formulated within this project to analyze the period characteristics of a song and its structural evolution over time:

- RQ1 (Machine Learning Task): *To what extent can a song’s release decade be predicted using Spotify audio features and Genius lyric-complexity features in a multi-class classification setting?*
- RQ2 (Hypothesis Testing): *Is there a statistically significant difference between older and modern songs in terms of their mean “acousticness” and “energy” values?*

This initial study performs data profiling, incomplete data analysis, and descriptive statistics on the selected *SpotGenTrack* dataset to conduct the necessary data quality checks for the research questions. Furthermore, a naïve baseline model is established for the targeted machine learning task, and appropriate performance evaluation metrics are defined, thus preparing a solid foundation for the final phase of the project.

II. DATA PROFILING AND DESCRIPTIVE STATISTICS

A. Selected Features

To give answers to the two proposed research questions, a set of audio, lyric complexity, and metadata variables were selected. For the machine learning part of the project, Spotify audio features like “acousticness”, “danceability”, “energy”, “instrumentalness”, “liveness”, “speechiness”, “tempo” and “valence” and variables such as “mean_syllables_word”, “mean_words_sentence”, “n_sentences”, “n_words”, “sentence_similarity” and “vocabulary_wealth” which are related to lyric complexity were chosen as independent variables. Additionally, in order to represent artist-level information, the variables called “artist_popularity” and “followers” were included into analysis by degrading to song

level. For this task, the target variable is the “*decade*” variable which was derived from the “*release_year*” variable.

For the hypothesis testing task, the main variables of focus were chosen as “*acousticness*” and “*energy*”. The songs were separated into two groups according to their release date:

- Older songs: Songs that were released either in 1980s or before fall into this group.
- Modern songs: Songs that were released either in 2010s or later fall into this group.

The purpose of this grouping was to investigate whether songs that are from distant musical periods differ in terms of these two audio characteristics or not.

B. Descriptive Statistics

Descriptive statistics were computed for the variables that are most relevant to the selected research questions. These variables include the selected audio definers and lyrics complexity measures. The results represent that the chosen variables are differing from each other significantly in terms of scale and dispersion. For instance, variables such as “*acousticness*”, “*danceability*”, and “*energy*” are bounded between 0 and 1, whereas variables such as “*tempo*” and “*n_words*” are falling into much larger intervals. This situation shows that the chosen attributes represent different aspects of a song and shows that a careful preprocessing is necessary in the modelling phase.

A summary statistics table related to the chosen variables are shown below. Additionally, to strengthen and to support the second research question, group-level descriptive statistics were calculated for “*acousticness*” and “*energy*” across the older and modern song categories.

TABLE I. DESCRIPTIVE STATISTICS OF SELECTED FEATURES

Feature	Count	Mean	Standard Deviation
acousticness	101939.0	0.352	0.335
danceability	101939.0	0.586	0.178
energy	101939.0	0.586	0.26
tempo	101939.0	118.36	30.224
mean_syllables_word	79420.0	1.456	0.368
mean_words_sentence	79420.0	4.488	10.296
n_words	79420.0	379.08	940.457
vocabulary_wealth	79420.0	0.557	0.12

C. Data Quality Checks

Various data quality checks were conducted before the analysis. Firstly, in order to merge the data from different tables a common song identifier, a key, was standardized. The variable called “*id*” on the Spotify track-level table, was renamed into “*track_id*” in order to make it consistent with the naming on the other tables related with albums, artists and lyrics. In addition, the columns that have no value such as “*Unnamed: 0*” and which act only as indexes were removed before analysis.

Secondly, the logs that were constantly repeating were analyzed. Especially since some songs on the artists table are related with more than one artist, it was observed that there are more than one rows which have the same “*track_id*” value. In order to create one song-level table for analysis, artist

information was grouped on the “*track_id*” basis and one row for per song was left. In this study, maximum values for the “*artist_popularity*” and “*followers*” variables were preserved.

Thirdly, the “*release_date*” variable was transformed into a date format, and “*release_year*” variable was obtained from that. By using this variable, for the machine learning task, the categorical target variable called “*decade*” and the group variable named “*era_group*” for the hypothesis testing were created.

Finally, class distribution was analyzed in terms of classification task. The resulting distribution shows that recent decades contain substantially more tracks than earlier decades. This imbalance is visualized in Fig.1 and has direct implications for metric selection in the machine learning phase.

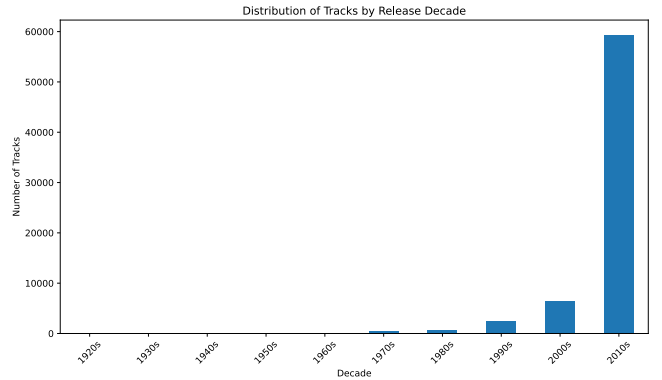


Fig.1. Distribution of Tracks by Release Date

D. Missing Values and Final Report Plan

As a part of data profiling phase, the missing and invalid values were also examined. It was observed that in the table that includes lyrics features, some variables have the “-1” value and it was concluded that these are not real measurements. They were used as a replacement to represent the missing information. Thus, through Python, the related values were recoded as “*NaN*” before descriptive analysis and data preparation. After merging the tables, it was observed that there are still missing values existent in some lyric and metadata variables.

For this interim report phase, these missing values were documented, and a complete-case filtering approach was applied where necessary to prepare the machine learning dataset. However, this approach is not the only possible solution. In the final report, alternative strategies such as selective imputation, exclusion of certain variables, or complete-case analysis will be evaluated depending on the amount and distribution of missing data. Additionally, some decade classes with very few observations in the interim dataset created for the machine learning task were excluded during the baseline evaluation. This is because the stratified train-test split requires a certain minimum number of observations for each class, and excessively sparse classes do not produce a reliable evaluation.

III. PERFORMANCE METRICS AND NAÏVE BASELINE MODEL

A. Selected Evaluation Metrics

Since the machine learning task is a multi-class classification problem with an imbalanced decade distribution, raw accuracy alone is not sufficient to evaluate

performance. Therefore, macro-F1 and balanced accuracy were chosen as the main performance metrics. Macro-F1 gives equal importance to each class by taking the average of F1 scores across decades, which makes it suitable when minority classes are present. Balanced accuracy is also appropriate because it averages recall across classes and therefore reduces the dominance of the largest class.

In addition, the accuracy value will be reported as a complementary metric. However, the main limitation of accuracy in this study is that it can appear artificially high when the model predicts the dominant decade most of the time. Thus, when comparing the machine learning models, it is more reasonable to choose macro-F1 and balanced accuracy metrics.

B. Naïve Baseline Model

A naïve baseline was defined using a majority-class classifier. It predicts the most frequent decade for every observation in the test set and does not require any training beyond identifying the dominant class in the training data. Therefore, this provides a suitable reference point for this interim report. Models such as Random Forest and XGBoost, which will be applied in the final phase, are expected to surpass this simple reference level in order to be considered meaningful.

The strongest aspect of this baseline is the fact that it is simple and interpretable. It provides a minimum performance level to evaluate whether the machine learning model is indeed beneficial or not. On the contrary, the weakest aspect of it is the fact that it does not use any feature in the dataset and performs poorly especially with the classes that have very few observations. Thus, it is expected to produce relatively weak macro-F1 and balanced accuracy values even if its accuracy is not trivial.

C. Baseline Results

The majority class baseline model was evaluated using the same metrics as those used in the final stage: accuracy, macro-F1, and balanced accuracy. The results are presented in Table 2 below. As expected, this model provides only a lower bound reference for the release decade classification task. While the accuracy value is able to reach a certain level because the model consistently predicts the dominant decade class, it is expected to yield weak results in terms of macro-F1 and balanced accuracy. This suggests that using imbalance-sensitive metrics instead of raw accuracy is more appropriate for this problem with class imbalance.

TABLE II. PERFORMANCE OF THE MAJORITY-CLASS NAÏVE BASELINE FOR RELEASE DECADE

<i>Metric</i>	<i>Naïve Baseline Score</i>
Accuracy	0.8662057044079516
Macro-F1	0.13261525432699942
Balanced Accuracy	0.14285714285714285

The majority class baseline results given in Table 2 above provide a key benchmark for the release decade classification task. The performance of the Random Forest and XGBoost models to be applied in the final report will be compared with these baseline results by using the same metrics.

IV. CONCLUSION

This interim report outlines the necessary preparations for the final analyses to be conducted on the *SpotGenTrack* dataset. Two research questions have been clarified, relevant variables were selected, data cleaning and merging steps were completed, and data quality checks were assessed using descriptive analysis. Furthermore, performance metrics and a pure baseline approach to be used for the release era/decade classification task have been determined. Initial observations indicate that the dataset has an unbalanced structure, particularly in terms of release decades, and that some lyric variables contain missing or invalid values. In the final report, hypothesis testing will be conducted based on these findings, followed by machine learning analysis using models such as Random Forest and XGBoost to seek more comprehensive answers to the research questions.

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